

5-1e Total Revenue and the Price Elasticity of Demand

When studying changes in supply or demand in a market, one variable we often want to study is **total revenue**, the amount paid by buyers and received by sellers of a good. In any market, total revenue is $P \times Q$, the price of the good times the quantity of the good sold. We can show total revenue graphically, as in [Figure 2](#). The height of the box under the demand curve is P , and the width is Q . The area of this box, $P \times Q$, equals the total revenue in this market. In [Figure 2](#), where $P = \$4$ and $Q = 100$, total revenue is $\$4 \times 100$, or $\$400$.

Figure 2

Total Revenue

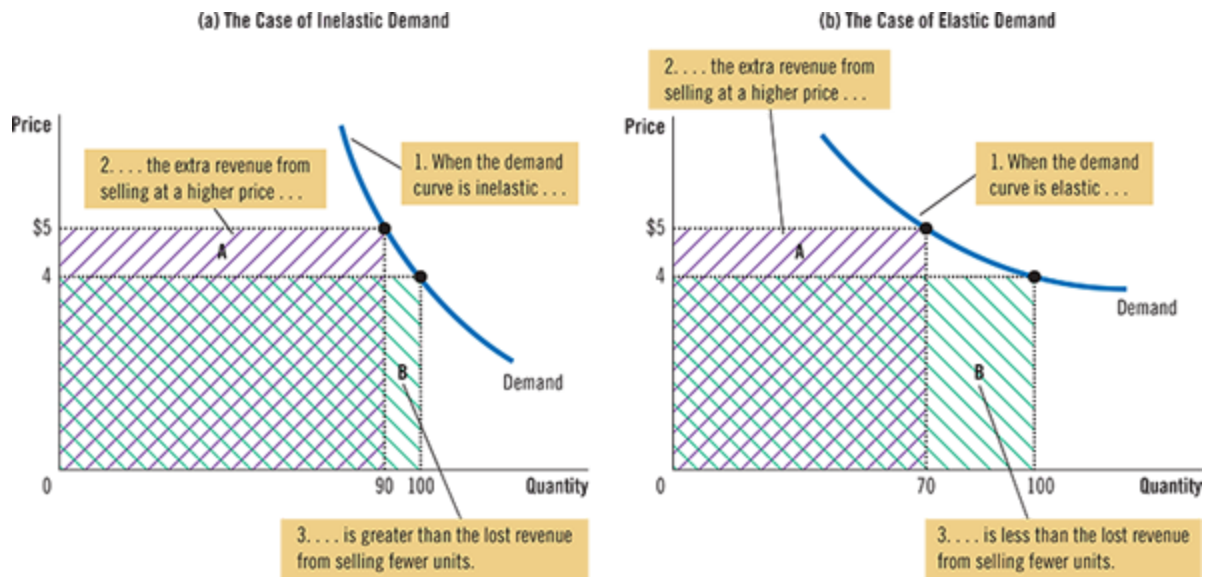
The area of the box under the demand curve, $P \times Q$, equals the total amount paid by buyers as well as the total revenue received by sellers. Here, at a price of $\$4$, the quantity demanded is 100 and total revenue is $\$400$.

How does total revenue change as one moves along the demand curve? The answer depends on the price elasticity of demand. If demand is inelastic, as in panel (a) of [Figure 3](#), then an increase in the price causes an increase in total revenue. Here an increase in price from $\$4$ to $\$5$ causes the quantity demanded to fall from 100 to 90, so total revenue rises from $\$400$ to $\$450$. An increase in price raises $P \times Q$ because the fall in Q is proportionately smaller than the rise in P . In other words, the extra revenue from selling units at a higher price (represented by area A in the figure) more than offsets the decline in revenue from selling fewer units (represented by area B).

Figure 3

How Total Revenue Changes When Price Changes

The impact of a price change on total revenue (the product of price and quantity) depends on the elasticity of demand. In panel (a), the demand curve is inelastic. In this case, an increase in the price leads to a proportionately smaller decrease in quantity demanded, so total revenue increases. Here an increase in the price from \$4 to \$5 causes the quantity demanded to fall from 100 to 90. Total revenue rises from \$400 to \$450. In panel (b), the demand curve is elastic. In this case, an increase in the price leads to a proportionately larger decrease in quantity demanded, so total revenue decreases. Here an increase in the price from \$4 to \$5 causes the quantity demanded to fall from 100 to 70. Total revenue falls from \$400 to \$350.



We obtain the opposite result if demand is elastic: An increase in the price causes a decrease in total revenue. In panel (b) of Figure 3, for instance, when the price rises from \$4 to \$5, the quantity demanded falls from 100 to 70, so total revenue falls from \$400 to \$350. Because demand is elastic, the reduction in the quantity demanded is so great that it more than offsets the increase in the price. That is, an increase in price reduces $P \times Q$ because the fall in Q is proportionately greater than the rise in P . In this case, the extra revenue from selling units at a higher price (area A) is smaller than the decline in revenue from selling fewer units (area B).

The examples in this figure illustrate some general rules:

- When demand is inelastic (a price elasticity less than one), price and total revenue move in the same direction: If the price increases, total revenue also increases.
- When demand is elastic (a price elasticity greater than one), price and total revenue move in opposite directions: If the price increases, total revenue decreases.

- If demand is unit elastic (a price elasticity exactly equal to one), total revenue remains constant when the price changes.

Chapter 5: Elasticity and Its Application: 5-1e Total Revenue and the Price Elasticity of Demand

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